Finance and Fintech Roadmap

One sequence for domain experience as a software engineer, investing, fintech roles, and real estate

Goals

Fintech roles	Domain knowledge and a portfolio project that clear bank and fintech interview screens.
Investing	Enough theory, tax knowledge, and practice to run your own long-term portfolio.
Real estate	The finance tools to evaluate a property and build your own underwriting model.
AI crossover	Quant and ML depth that pairs with your McDaniel MS in Data Analytics.

How to use this. It assumes about 5 hours a week alongside job searching and McDaniel coursework. Stages 1 to 5 take roughly 10 months; Stage 6 continues alongside your degree. Your five years in backend and banking systems mean you are filling in domain knowledge, not starting from zero, so audit courses where you can and run lectures at 2x on material you already know. Platform availability changes, so confirm each course is still listed when you reach it.

STAGE 1 — Shared Foundation

Weeks 1-5 | ~25 hrs

Every roadmap you uploaded starts here. Accounting is the language everything else is written in, and you need just enough personal finance to follow tax and credit concepts later.

Introduction to Financial Accounting

UPenn / Wharton - Prof. Brian Bushee - Coursera (audit free) - ~20 hrs

Balance sheet, income statement, cash flow, accrual accounting. Retroactively explains what the ledger and reconciliation systems you built were doing.

Khan Academy: Financial Literacy (Units 3, 10, 12 only)

Khan Academy - free - ~3 hrs

Consumer credit, taxes, and banking. Skip the other units; you don't need them for these goals.

Outcome: You can read a financial statement and follow credit, tax, and banking vocabulary without stopping to look things up.

STAGE 2 Fintech Domain Knowledge

Weeks 6-15 | ~50 hrs

The vocabulary and plumbing that bank and fintech interviews assume. This is the stage that turns your Morgan Stanley experience into a financial story, not just a technical one.

Financial Markets

Yale - Prof. Robert Shiller - Coursera (audit free) - ~33 hrs, ~20 at 2x speed

Equities, bonds, options and futures, banking, regulation, 2008. Prioritize the bonds/options and banking/exchange modules.

Economics of Money and Banking

Columbia - Prof. Perry Mehrling - Coursera (audit free) - ~20 hrs

Bank balance sheets, repos, money markets, SOFR, dealer banks, shadow banking. The overnight plumbing of finance.

Trade Lifecycle and Surveillance Reading Sprint

Investopedia, SEC, FINRA, industry articles - free - ~10 hrs

Execution to clearing to settlement to custody (T+1), prime brokerage, spoofing, layering, wash trading, front-running, SEC/FINRA/CFTC roles, Dodd-Frank, MiFID II.

Outcome: Fluent market vocabulary and concrete interview talking points on trade lifecycle, surveillance, and regulation.

STAGE 3 Portfolio Project

Weeks 16-19 | ~20 hrs

Placed before the investing material on purpose: this is the most leveraged thing you can show a fintech hiring manager while you are still job searching.

Build: Trade Surveillance Signal Detector

Personal project - GitHub - Java or Python

Ingest a public market dataset, implement 2-3 detection rules (spoofing, wash trading, volume spikes near close), expose a REST API that returns flagged events, add a simple output view, and write a README explaining the finance behind each rule.

Outcome: A GitHub project that shows you understand what surveillance and risk systems actually do.

STAGE 4 Investing

Weeks 20-32 | ~65 hrs

Practical investing plus the tax and portfolio theory behind it. Real money makes the material stick, so consider opening a brokerage account early in this stage and putting a small recurring amount into a broad index fund. What to buy is your decision, and this is not financial advice.

Complete Investing Course 2026 (Stocks, ETFs and Funds)

Udemy via library card - Calvin K. Lee, CPA/CA/MBA - ~8 hrs

Account types, ETFs vs mutual funds vs REITs, passive vs active, building a simple portfolio.

Federal Taxation I: Individuals, Employees and Sole Proprietors

Univ. of Illinois - Coursera (Plus) - ~30 hrs

Brackets, W-2 vs 1099, deductions, retirement accounts (IRA, Roth, 401k), a hands-on Form 1040. After-tax returns drive every investing decision.

Investment and Portfolio Management Specialization (courses 1-2)

Rice University - Prof. Arzu Ozoguz - Coursera (Plus) - ~30 hrs

Risk and return, diversification, CAPM, mean-variance optimization. Stanford GSB's free Stocks and Bonds course is a lighter substitute.

Optional: Practical Financial Statement Analysis

Udemy via library card - Jason Williams - ~10 hrs

Reading 10-Ks, key ratios, red flags. Useful if you plan to pick individual stocks.

Outcome: You can open and run a sensible long-term portfolio and understand its tax consequences.

STAGE 5 Real Estate

Weeks 33-44 | ~60 hrs

The weakest area in the source material: your uploads covered real estate with one Alison course and a placeholder. The plan below fills it with corporate finance fundamentals and a model you build yourself.

Corporate Finance Essentials

IESE - Prof. Javier Estrada - Coursera (Plus) - ~13 hrs

NPV, IRR, DCF, WACC, cost of capital. These are the tools underneath every property analysis.

Introduction to Real Estate Investing

Alison - free - ~8 hrs

Mortgages, home equity, REITs, cap rates, property valuation basics.

Real Estate Economics and Financial Modeling

Course to be chosen when you reach this stage - ~25 hrs

Housing supply and demand, interest-rate effects, pro formas, amortization, and cash-flow DCF for a property. Your spreadsheet only listed 'LSE or open equivalent', so pick a specific course then.

Capstone: Underwrite a Real Listing

Personal project - Python or Excel - ~15 hrs

Model purchase price, financing, rent, expenses, vacancy, and returns for a real listing, including rent-vs-buy and downside cases.

Outcome: You can evaluate a property the way a lender or investor would, with a model you built.

STAGE 6 Fintech Engineering Depth

Weeks 45+ | ~180 hrs

Where finance and your McDaniel AI track reinforce each other. This stage runs alongside coursework and can be reordered to match what you are studying that semester.

Financial Engineering and Risk Management Specialization

Columbia - Profs. Haugh and Iyengar - Coursera - 5 courses, ~60 hrs

Options, Black-Scholes, Greeks, swaps, credit derivatives, portfolio optimization, VaR. Aim to understand what the instruments are and why firms use them.

Financial Technology (Fintech) Innovations Specialization

Univ. of Michigan - Coursera (Plus) - 4 courses

Payment systems, blockchain, crowdfunding, AI in robo-advising.

Python for Finance and Algorithmic Trading with QuantConnect

Udemy via library card - Jose Portilla

NumPy, pandas, yfinance, backtesting real strategies.

Machine Learning for Trading Specialization

Google Cloud and NYIF - Coursera (Plus)

Take only after a McDaniel ML course. Pairs well with applied projects like transaction anomaly detection or credit default prediction.

Outcome: Technical depth in derivatives, risk, and quant methods, plus portfolio pieces that combine ML with finance.

Timeline Summary

Stage	Weeks	Focus	Main courses	Hours
1	1-5	Accounting and money basics	Wharton Accounting; Khan (3 units)	~25 hrs
2	6-15	Markets, bank plumbing, regulation	Shiller; Mehrling; reading sprint	~50 hrs
3	16-19	Surveillance detector project	Java or Python + REST API	~20 hrs
4	20-32	Investing, taxes, portfolio theory	Udemy Investing; Taxation I; Rice 1-2	~65 hrs
5	33-44	Property analysis and modeling	IESE Corp Finance; Alison RE; RE modeling	~60 hrs
6	45+	Derivatives, fintech, ML for finance	Columbia FERM; Michigan Fintech; QuantConnect; ML for Trading	~180 hrs

What Was Cut, and Why

These courses appeared in your uploads and are fine, but none moves you toward fintech roles, investing, or real estate faster than the stages above: Duke Behavioral Finance, Managerial Accounting, Mergers and Acquisitions, Successful Negotiation, Operations Management, Darden Business Strategy, Alison's debt management course, and the 17-hour Udemy stock-picking course. Revisit them later if a specific role or interest calls for them.

Optional Extras

FINRA Series 99 (Operations Professional exam) signals commitment for capital markets operations roles. **CFA Level 1** makes sense only if you later move toward quant or product roles. For reading alongside the stages: *A Random Walk Down Wall Street* during Stage 4 and *The Big Short* after Stage 2 for 2008 context.

Educational planning material, not financial advice. Course details, prices, and availability should be verified on each platform.